

KoFEM — Meshing Capabilities

Surface tessellation test suite — original geometries — May 16, 2026

Geometry	Rendered surface	Surface mesh (wireframe)	Volume mesh (tets)
Simple Box 80 × 60 × 40 mm — 6 planar faces	screenshot missing	screenshot missing	screenshot missing
Cylinder R = 25 mm, H = 80 mm — cylindrical surface + 2 disc caps	screenshot missing	screenshot missing	screenshot missing
Truncated Cone R_bot = 10, R_top = 20, H = 30 mm — conical surface + 2 disc caps	screenshot missing	screenshot missing	screenshot missing
L-Bracket 80 × 80 × 20 mm — 8 planar faces, re-entrant corner	screenshot missing	screenshot missing	screenshot missing
Quarter-Cyl. Patch r = 5, H = 10 mm, u " partial cylindrical surface	screenshot missing	screenshot missing	screenshot missing
Complex Bracket Imported STEP — curved faces, blends, multiple features	screenshot missing	screenshot missing	screenshot missing

KoFEM — New Test Shapes

Tubes, bends, toroidal surfaces, prismatic profiles — May 16, 2026

Geometry	Rendered surface	Surface mesh (wireframe)	Volume mesh (tets)
Hollow Tube R ϕ = 20 , R b = 14 , H = 60 mm — annular caps with FACE_BOUND holes	screenshot missing	screenshot missing	screenshot missing
90° Pipe Elbow R $\sim$ = 40 , r = 10 mm — quarter TOROIDAL_SURFACE + 2 disc caps	screenshot missing	screenshot missing	screenshot missing
Half-Torus U-Bend R $\sim$ = 30 , r = 10 mm — Å-s TOROIDAL_SURFACE	screenshot missing	screenshot missing	screenshot missing
Stepped Shaft R 20 / 12 , H 30 + 40 mm — cylinders + annular step ring	screenshot missing	screenshot missing	screenshot missing
Hex Prism circumR=25, H=50 mm — regular hexagonal cross-section	screenshot missing	screenshot missing	screenshot missing
Square Pyramid base 50×50 mm, H=60 mm — 4 triangular faces + square base	screenshot missing	screenshot missing	screenshot missing
Triangular Wedge 80×50×30 mm — right-angle triangular prism	screenshot missing	screenshot missing	screenshot missing
I-Beam W=60, H=80, tf=8, tw=6, L=80 mm — structural steel section	screenshot missing	screenshot missing	screenshot missing

KoFEM — New Test Shapes

Tubes, bends, toroidal surfaces, prismatic profiles — May 16, 2026 (continued 2/2)

Geometry	Rendered surface	Surface mesh (wireframe)	Volume mesh (tets)
T-Profile W=80, H=68, tf=8, tw=10, L=20 mm — T cross-section extrusion	screenshot missing	screenshot missing	screenshot missing
U-Channel W=60, H=40, t=5, L=80 mm — C/U section extrusion	screenshot missing	screenshot missing	screenshot missing

KoFEM — NIST AP242 Standard Cases

NIST MBE PMI Validation & Conformance Testing — May 16, 2026

Geometry	Rendered surface	Surface mesh (wireframe)	Volume mesh (tets)
NIST CTC-01 AP242 e1 — Core Technical Case 1	screenshot missing	screenshot missing	screenshot missing
NIST CTC-02 AP242 e2 — Core Technical Case 2	screenshot missing	screenshot missing	screenshot missing
NIST CTC-03 AP242 e2 — Core Technical Case 3	screenshot missing	screenshot missing	screenshot missing
NIST CTC-04 AP242 e1 — Core Technical Case 4	screenshot missing	screenshot missing	screenshot missing
NIST CTC-05 AP242 e1 — Core Technical Case 5	screenshot missing	screenshot missing	screenshot missing
NIST FTC-06 AP242 e2 — Feature Technical Case 6	screenshot missing	screenshot missing	screenshot missing
NIST FTC-07 AP242 e2 — Feature Technical Case 7	screenshot missing	screenshot missing	screenshot missing
NIST FTC-08 (tg) AP242 e1 — FTC-08 tolerance geometry	screenshot missing	screenshot missing	screenshot missing

KoFEM — NIST AP242 Standard Cases

NIST MBE PMI Validation & Conformance Testing — May 16, 2026 (continued 2/3)

Geometry	Rendered surface	Surface mesh (wireframe)	Volume mesh (tets)
NIST FTC-08 AP242 e2 — Feature Technical Case 8	screenshot missing	screenshot missing	screenshot missing
NIST FTC-09 AP242 e1 — Feature Technical Case 9	screenshot missing	screenshot missing	screenshot missing
NIST FTC-10 AP242 e2 — Feature Technical Case 10	screenshot missing	screenshot missing	screenshot missing
NIST FTC-11 AP242 e2 — Feature Technical Case 11	screenshot missing	screenshot missing	screenshot missing
NIST STC-06 AP242 e3 — Sheet Technical Case 6	screenshot missing	screenshot missing	screenshot missing
NIST STC-07 AP242 e3 — Sheet Technical Case 7	screenshot missing	screenshot missing	screenshot missing
NIST STC-08 AP242 e3 — Sheet Technical Case 8	screenshot missing	screenshot missing	screenshot missing
NIST STC-09 AP242 e3 — Sheet Technical Case 9	screenshot missing	screenshot missing	screenshot missing

KoFEM — NIST AP242 Standard Cases

NIST MBE PMI Validation & Conformance Testing — May 16, 2026 (continued 3/3)

Geometry	Rendered surface	Surface mesh (wireframe)	Volume mesh (tets)
NIST STC-10 AP242 e2 — Sheet Technical Case 10	screenshot missing	screenshot missing	screenshot missing